Bias and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Recall and choose your starting point

> Record your starting explanation for Bias and Representation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Copyright and Provenance

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Red Team the Product

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prototype a Grounded Assistant

Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Bias and Representation trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Representation** and bias evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Bias and Representation explanation against the supplied facts before discussing it.

Use the worked case for **Representation** and bias evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Bias and Representation target to a trace and a justified conclusion.

An illustration set always shows scientists in the same role and appearance. Define a review that examines **representation** without inferring personal identities from images.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Bias and Representation case and show the deciding evidence.

Scientist images repeatedly show only one demographic group.

Evaluate and improve **representation**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

An assistant is tested on six fluent requests but no requests with spelling errors. What gap does this leave, and how should the evaluation expand?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Bias and Representation adds to Verified Study Buddy and keep the evidence.

Audit answer quality across reading levels, languages, names, and accessibility needs.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

An illustration set always shows scientists in the same role and appearance. Define a review that examines **representation** without inferring personal identities from images.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Bias and Representation facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Bias and Representation facts and keep your reasoning.

A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Bias and Representation facts and keep your reasoning.

A report omits this qualification: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Bias and Representation facts and keep your reasoning.

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Bias and Representation facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Bias and Representation

> Compare study-answer support rates when groups have different numbers of questions. Explain the labelled step before checking the changed condition.

Compare study-answer support rates when groups have different numbers of questions.

Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases. Both groups use the same support rubric.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/30

Bias and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Keep denominators

Birch: 6/8. Pine: 9/12.

↓

2. Compare rates

Both support rates are 75%; Pine has more correct answers because it has more cases.

↓

3. Inspect errors

Check the two Birch and three Pine failures before deciding what revision each needs.

CHANGE ONE THING

Pine instead has 6 supported answers out of 12.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare study-answer support rates when groups have different numbers of questions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check the two Birch and three Pine failures before deciding what revision each needs.

1. Compare rates

2. Inspect errors

3. Keep denominators

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect errors Check the two Birch and three Pine failures before deciding what revision each needs. Read the picture in order.

Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Pine instead has 6 supported answers out of 12. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Pine is 50%, which is 25 percentage points below Birch.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Pine is 50%, which is 25 percentage points below Birch. Equal correct counts do not imply equal rates when denominators differ.

Repair a missing building block

Review Copyright and Provenance (ai-3/29). Goal: Explain the **mechanism** and terminology of copyright and permitted sources. Short practice: Does acknowledging Fern make a cropped version permitted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Prototype a Grounded Assistant (ai-3/24). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: Can PL2 alone support “check the soil daily”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

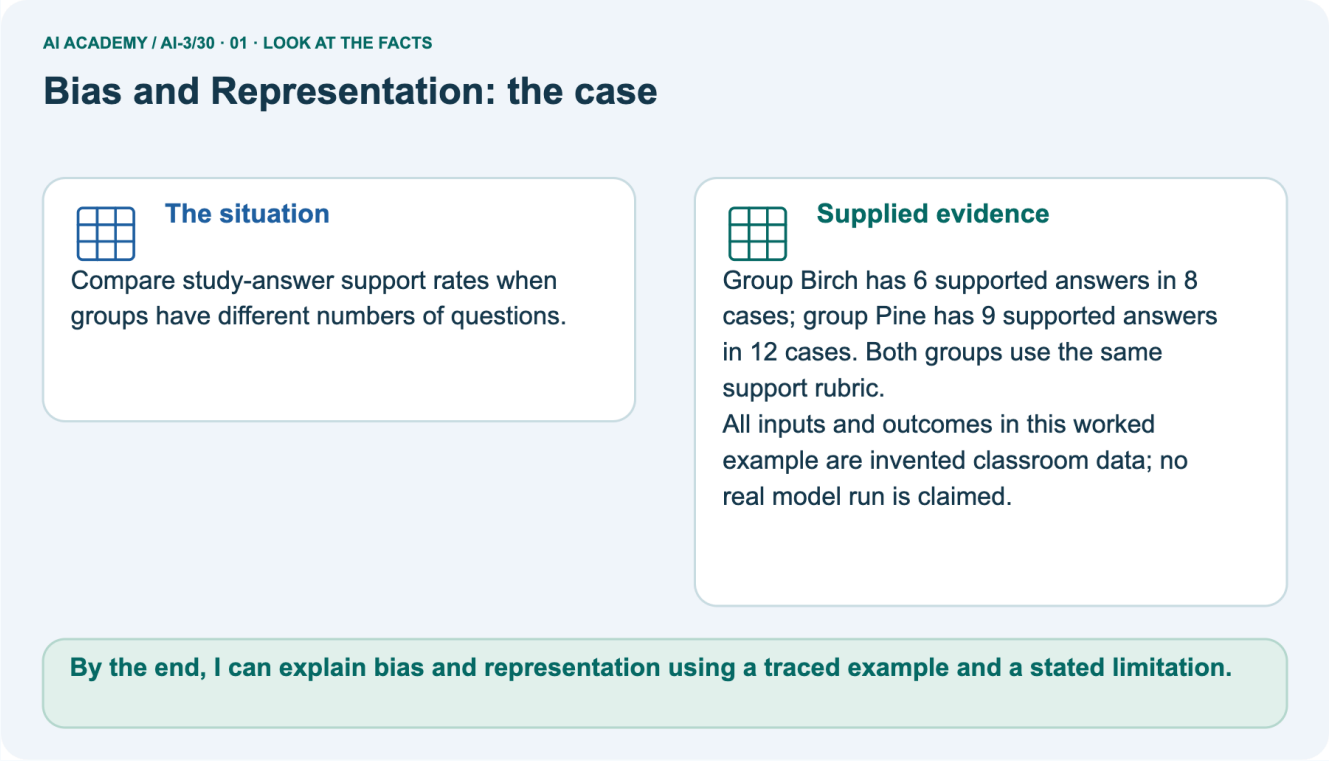
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Bias and Representation picture from its labels, then record a relevant limit.

01. Bias and Representation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

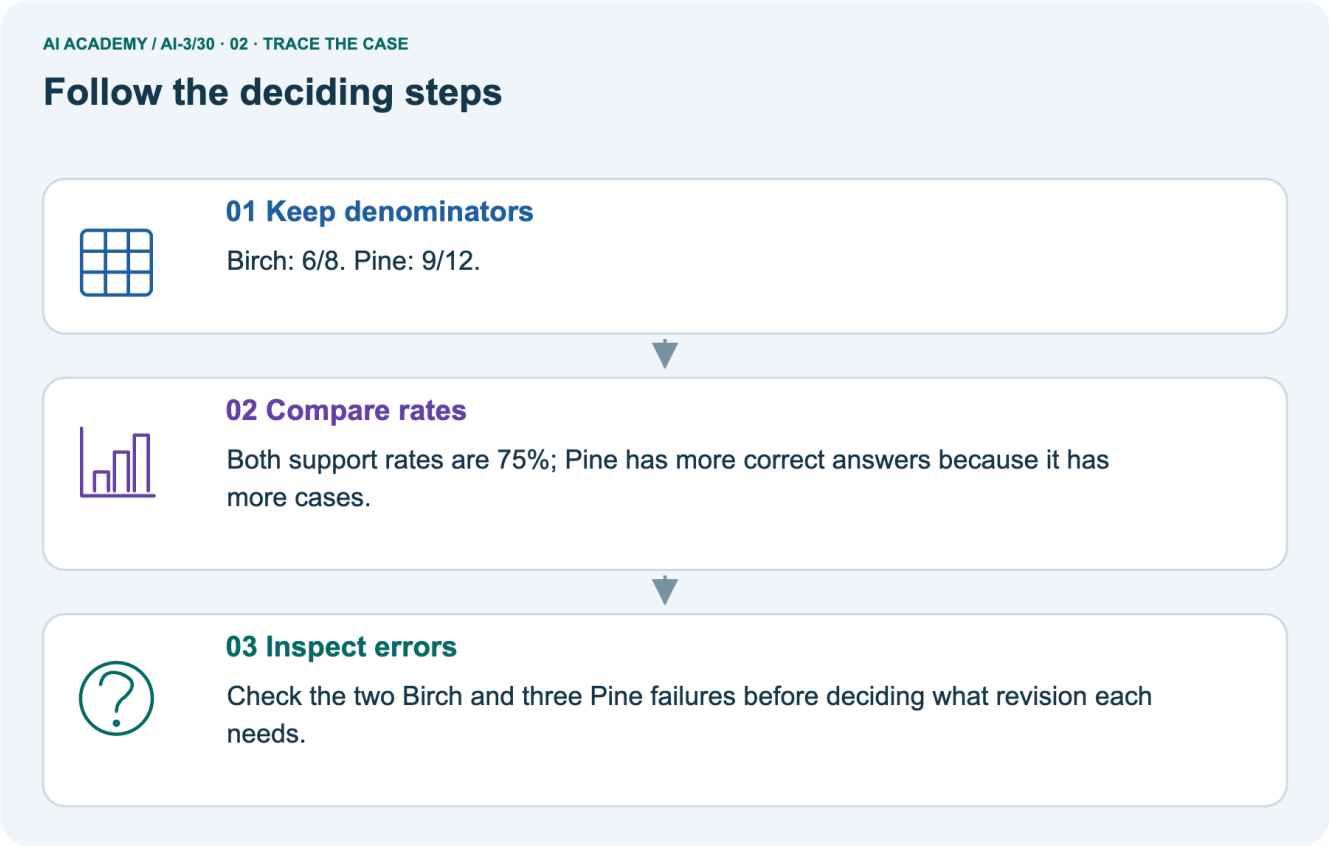
Compare study-answer support rates when groups have different numbers of questions. Supplied facts: Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases.

Both groups use the same support rubric. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

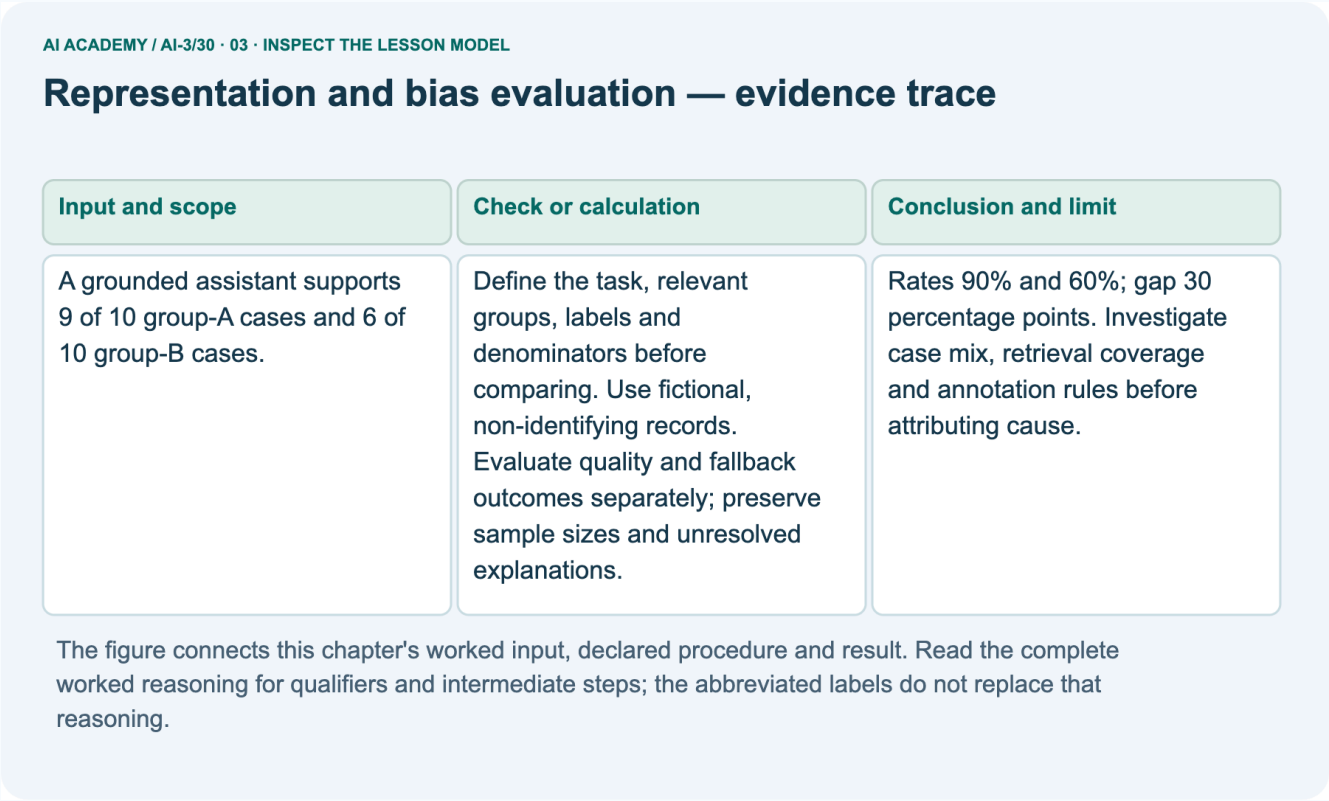
Keep denominators: Birch: 6/8.

Pine: 9/12. → Compare rates: Both support rates are 75%; Pine has more correct answers because it has more cases. → Inspect errors: Check the two Birch and three Pine failures before deciding what revision each needs.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Representation and bias evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Representation and bias evaluation — **evidence** trace. A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. | Define the task, relevant groups, labels and denominators before comparing.

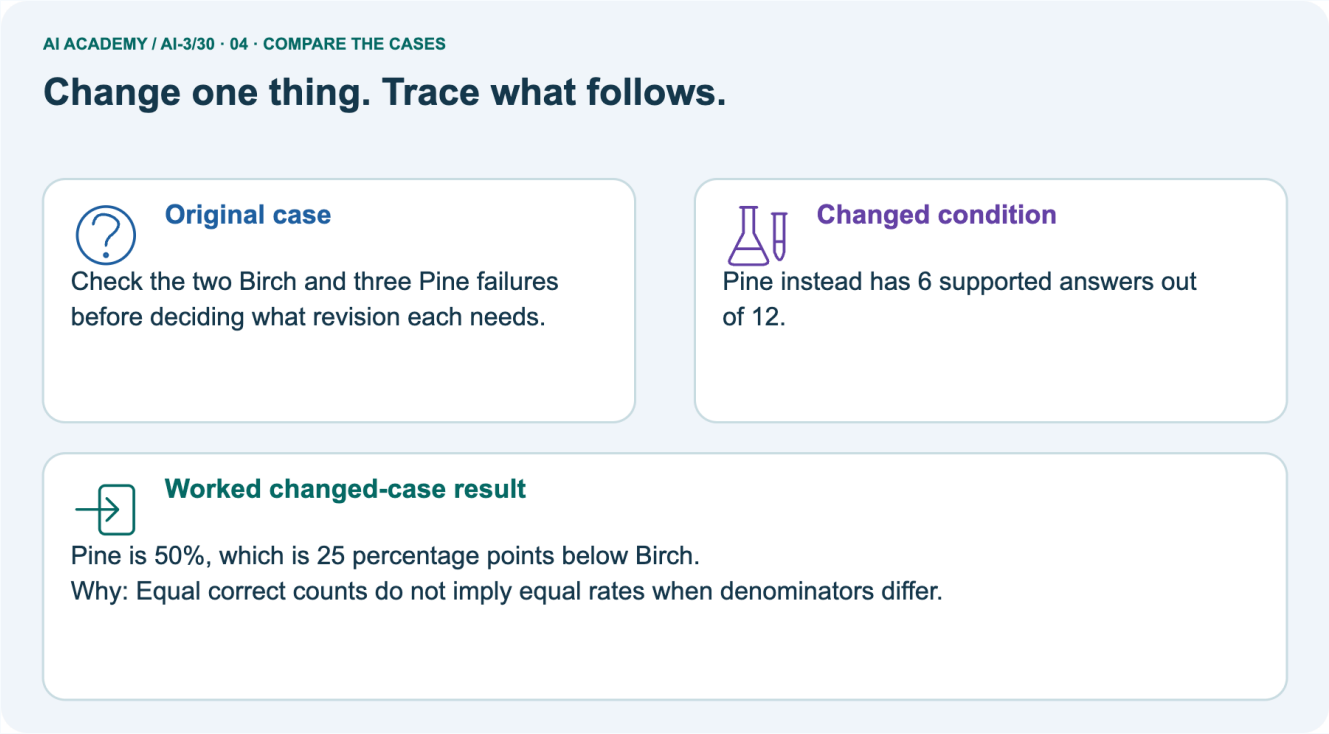
Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations. | Rates 90% and **60%**; gap 30 percentage points.

Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

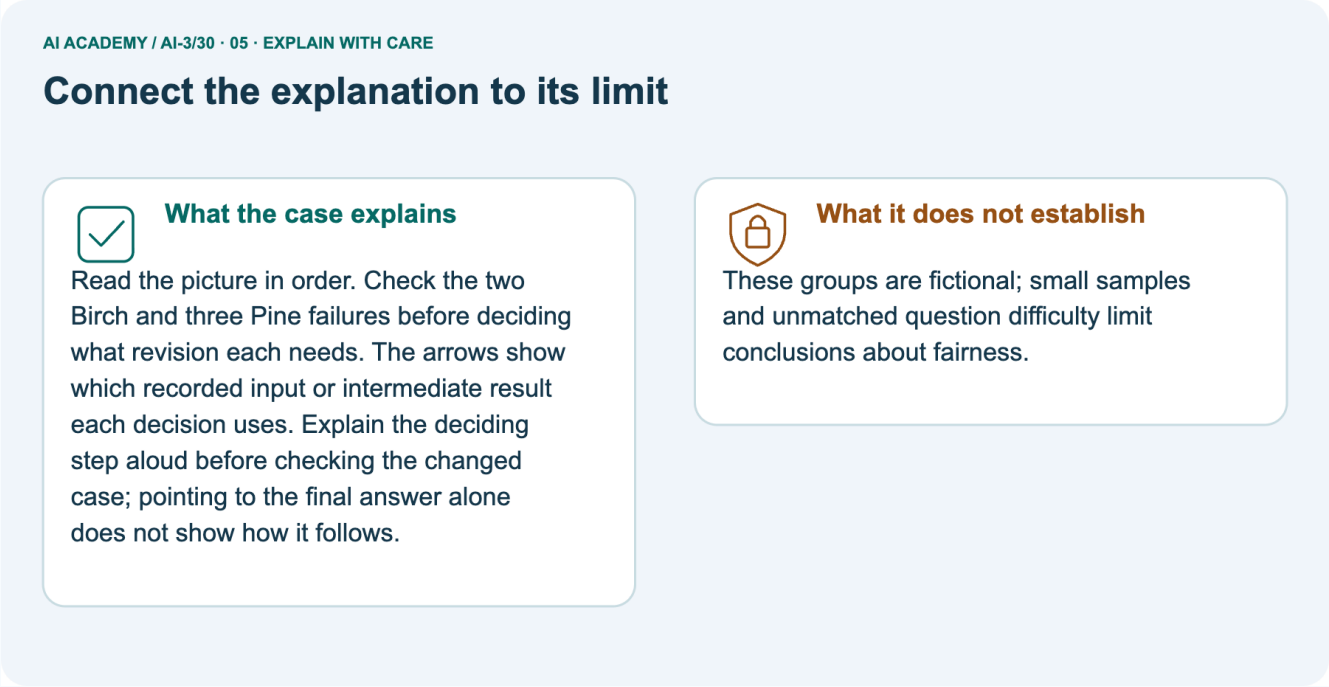
Original case: Check the two Birch and three Pine failures before deciding what revision each needs. Changed condition: Pine instead has 6 supported answers out of 12.

Expected result: Pine is 50%, which is 25 percentage points below Birch. Reason: Equal correct counts do not imply equal rates when denominators differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded **input** or intermediate result each decision uses.

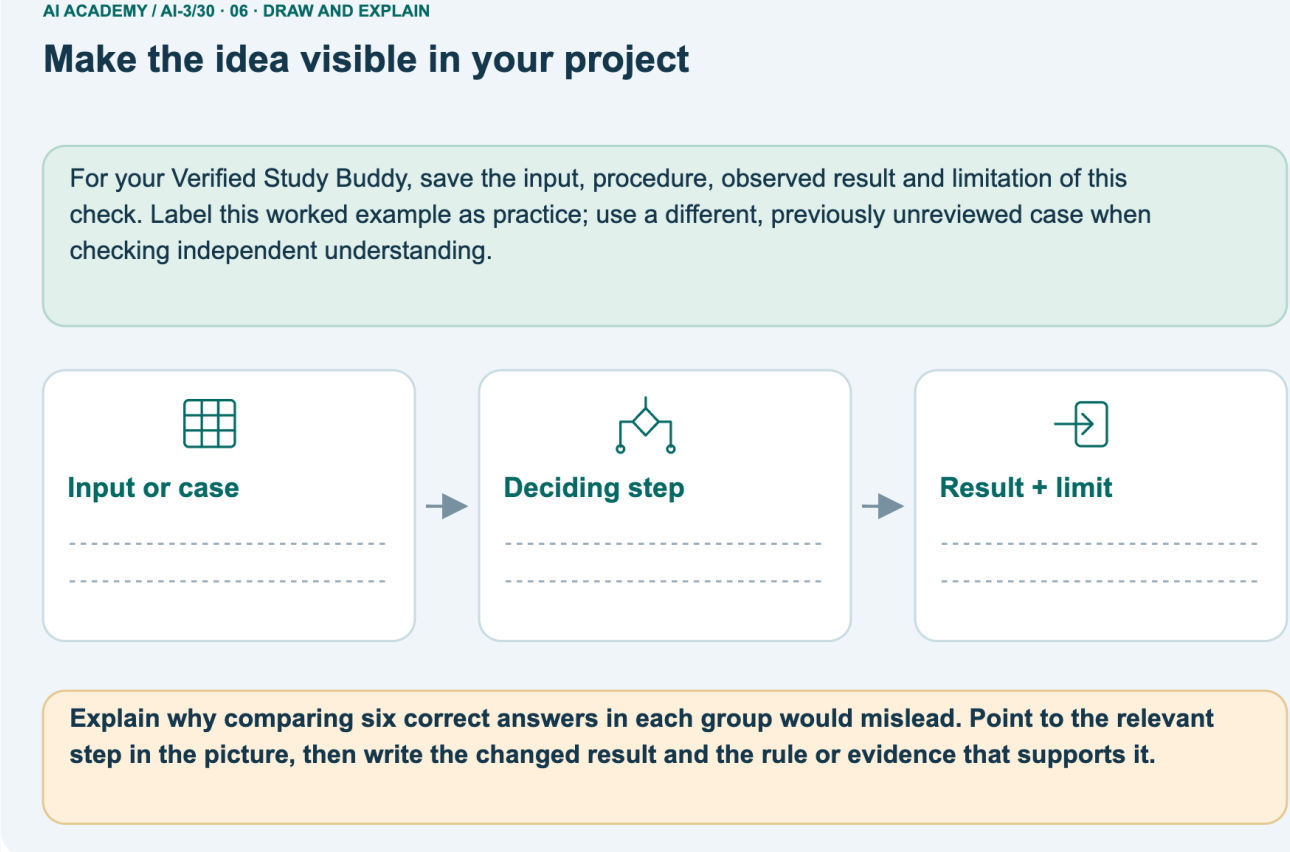
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These groups are fictional; small samples and unmatched question difficulty limit conclusions about fairness.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.